

Notes: Greenwood and Sammon (JF, 2025)

The Disappearing Index Effect

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1 Big picture

- **Question.** Why have the abnormal returns around S&P 500 additions and deletions nearly disappeared even though passive indexation has grown dramatically?
- **Classic benchmark.** Earlier papers such as Shleifer (1986) and Harris and Gurel (1986) found sizeable announcement effects when a stock entered the S&P 500. A simple inelastic-demand view says that if more dollars mechanically track the index, the index effect should get larger, not smaller.
- **Core puzzle.** The paper writes price impact as

$$\text{Price Impact}_{it} = M \times D_{it},$$

where D_{it} is the net mechanical demand shock created by index tracking and M is the price-impact multiplier, equal to minus one over the demand elasticity. Since D_{it} has risen over time, the disappearance of the index effect means either that D_{it} is mismeasured or that M has fallen sharply.

- **Main empirical fact.** Average S&P 500 addition returns rose from about 3.4% in the 1980s to 7.4% in the 1990s, then fell to 5.2% in the 2000s and to only 0.8% in the 2010s. Deletion returns follow the same pattern in reverse sign: -4.6% , -16.1% , -12.4% , and finally -0.6% .
- **What the paper investigates.** The authors organize the explanation into four candidate channels: (i) changing composition of additions and deletions, (ii) migrations between the S&P MidCap and the S&P 500, (iii) greater predictability and front running, and (iv) a genuine increase in the market's ability to provide liquidity around index changes.
- **Main conclusion.** Composition changes explain only a modest part of the decline. Migrations matter a lot, especially because MidCap indexing has grown. Front-running and predictability matter somewhat, but only partially. The dominant force is that the market has become far more elastic in absorbing index demand shocks.
- **Elasticity result.** After accounting for migrations, the implied multiplier M for additions falls from about 6.75 in 1995–1999 to 0.37 in 2010–2020; for deletions it falls from 10.76 to 0.70. That is an enormous increase in effective market elasticity.
- **Mechanism the authors favor.** Over time, repeated index-rebalancing demand has attracted liquidity provision, specialized trading desks, more coordinated execution, and institutional counterparties willing to take the other side of indexers' trades.
- **External validity.** The same broad pattern appears in other index families — Russell, S&P MidCap and SmallCap, and Nasdaq 100 — though the statistical strength is weaker outside the S&P 500.
- **Economic takeaway.** The paper reads as evidence for adaptive efficiency. A famous demand-based anomaly became well known, capital was organized to trade against it, and the anomaly largely disappeared on average.

2 Data and measurement (key objects)

2.1 Event sample

The core setting is S&P 500 index additions and deletions from 1980 to 2020. The authors assemble index changes from S&P publications and related sources, then apply filters to remove events contaminated by listings, delistings, acquisitions, acquirers, and other special cases that would mechanically confound returns around the event window.

At the decade level, the main return sample contains:

- 196 additions and 39 deletions in the 1980s,
- 134 additions and 56 deletions in the 1990s,
- 211 additions and 85 deletions in the 2000s,
- 153 additions and 87 deletions in the 2010s.

Interpretation. The paper is not built on a few well-known inclusions. It is a long-horizon event-study data set spanning the rise of indexing itself.

2.2 Identifying S&P 500 trackers

To estimate mechanical demand, the paper identifies mutual funds and ETFs that track the S&P 500 using Thomson S12 holdings data. The basic idea is behavioral rather than label-based: a fund is treated as a tracker if it repeatedly buys additions and sells deletions around index changes. The authors then aggregate holdings across these trackers.

The key trading variable is

$$Net\ Trading_{i,t} = 100 \times \frac{\Delta Shares\ Held_{i,(t+1,t-1)}}{Shares\ Outstanding_{i,t+1}},$$

where holdings are compared from the quarter before to the quarter after the index event.

Interpretation. This is the paper's empirical measure of the demand shock D . It is the fraction of shares that index trackers must absorb or unload because of the index change.

2.3 How large has indexing become?

Using mutual funds and ETFs alone, the paper estimates that S&P 500 trackers have grown from essentially zero in the early decades to roughly 7% of aggregate market capitalization in recent years. The authors view this as a lower bound because it excludes direct indexing and other shadow-indexing arrangements.

For the average event, net buying by trackers rises from nearly 0% of shares outstanding in the early 1990s to more than 6% for additions and nearly 8% for deletions by the late 2010s and 2020.

Interpretation. Mechanical demand got much bigger. That is why the disappearance of the index effect is so surprising.

2.4 Return measurement

The main event-study object is the cumulative abnormal return

$$CAR_{it} = R_{it} - R_{S\&P500,t}.$$

For the S&P 500, the paper studies three windows:

- **Announcement return:** from the day before to the day after the announcement.
- **Effective-date return:** from the day before to the day after implementation.
- **Total return:** from the day before announcement to the day after implementation.

The typical gap between announcement and implementation is about 4.8 trading days for additions and 5.8 for deletions.

Interpretation. The total return is the paper's preferred measure because it is designed to capture the full price impact from the time the change becomes known until the market completes the rebalance.

2.5 Migrations versus direct additions and deletions

A key distinction is whether an event is a **migration**. A migration occurs when a stock moves between the S&P MidCap 400 and the S&P 500. For example:

- a stock added to the S&P 500 and simultaneously dropped from the MidCap,
- or a stock deleted from the S&P 500 and added to the MidCap.

These cases involve offsetting passive flows:

- S&P 500 trackers buy when a stock enters the S&P 500,
- but MidCap trackers sell at the same time.

The authors therefore construct separate demand-shock measures for migrations and for direct changes.

Interpretation. Looking only at S&P 500 tracker demand overstates the net shock if another passive index is simultaneously taking the opposite side.

2.6 MidCap tracker size

Because MidCap holdings data are less complete, the paper identifies S&P MidCap trackers using fund names and return correlations. The resulting estimate is that MidCap-linked assets also grow sharply over time, reaching about 6% of MidCap capitalization by the end of the sample.

Interpretation. This makes migrations increasingly important over time. In the 1990s, an S&P 500 inclusion from the MidCap still generated net buying pressure; by the 2010s, that offset could be very large.

2.7 Control variables for composition

To test whether composition drives the disappearing effect, the paper studies four stock characteristics:

- abnormal turnover over the month before the event,
- firm size relative to total S&P 500 capitalization,
- Wurgler and Zhuravskaya’s (2002) arbitrage-risk measure,
- analyst coverage before the event.

Interpretation. These are designed to capture liquidity, scale of the demand shock, difficulty of arbitrage, and investor attention.

2.8 Predictability variables

To study front running, the paper measures cumulative abnormal returns well before announcement, including windows ending one day before the announcement that begin 10, 20, 50, and 100 trading days earlier. It also ranks non-S&P firms by market capitalization to see whether future inclusions have become more mechanically forecastable.

2.9 Liquidity and ownership variables

To understand why the market became more elastic, the paper looks at:

- value-weighted effective spreads from TAQ data,
- implementation shortfall from ITG/Virtu data,
- changes in 13F institutional ownership,
- changes in passive and active mutual fund ownership,
- and changes in shares outstanding around index events.

Interpretation. The idea is to move beyond “the effect got smaller” and ask *who* now absorbs index demand and *how* the trading environment changed.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline demand-shock logic

The paper starts from the reduced-form price-impact relation

$$Price\ Impact_{it} = M \times D_{it}. \tag{1}$$

Here, D_{it} is the net mechanical demand shock as a share of firm capitalization, and M is the price-impact multiplier.

Interpretation. If demand is inelastic, M is large in absolute value, so passive demand moves prices a lot. If the market is elastic, M is small and the same rebalancing demand has little effect on price.

3.2 Event-study abnormal return

The main return measure is

$$CAR_{it} = R_{it} - R_{S\&P500,t}. \quad (2)$$

For most of the paper, CAR_{it} is the abnormal return from one day before the announcement to one day after implementation.

Interpretation. This is not a permanent-value estimate. It is a price-impact estimate around the index-change episode.

3.3 Composition regression

To ask whether changing firm characteristics explain the decline, the paper estimates

$$CAR_{it} = b_1 Turn_{i,t-1} + b_2 Size_{i,t-1} + b_3 WZ_{i,t-1} + b_4 Cover_{i,t-1} + \sum_{k=1}^4 \gamma_k 1_{era=k} + e_{it}. \quad (3)$$

The decade dummies capture residual period effects after controlling for characteristics.

Interpretation. If the disappearing index effect were only a composition story, the decade coefficients should shrink sharply or disappear once these controls are added.

3.4 Demand decomposition with migrations

The paper then rewrites average price impact by decade as

$$\overline{CAR} = M \times \overline{D} = M \times (w \cdot \overline{D}_{Migrations} + (1 - w) \cdot \overline{D}_{NonMigrations}), \quad (4)$$

where w is the share of events that are migrations.

Interpretation. This is the cleanest decomposition in the paper. It says the observed average return can shrink either because the net shock is smaller, or because the multiplier M is smaller.

3.5 Controlled multiplier regression

To incorporate both composition and demand-shock differences, the paper estimates

$$CAR_{it} = b_1 Turn_{i,t-1} + b_2 Size_{i,t-1} + b_3 WZ_{i,t-1} + b_4 Cover_{i,t-1} + \sum_{k=1}^3 \gamma_k 1_{era=k} \times \overline{D}_{era=k} + e_{it}. \quad (5)$$

The coefficients γ_k are interpreted as decade-specific multipliers M .

Interpretation. This is the paper's main attempt to isolate a decline in price impact that is not just due to different firms, different migration shares, or different measured demand-shock size.

3.6 Extended return window

The authors also re-estimate the multiplier using a longer event window, from 20 trading days before the announcement to one day after implementation.

Interpretation. This is a robustness test against the front-running explanation. If the main effect has merely moved earlier in event time, the longer-window estimate should not fall as much as the short-window one.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the event-study design is informative

S&P index changes are useful because the mechanical trading by index funds is unrelated to the fundamentals of the stock being bought or sold at that exact moment. The stock is reclassified because of index membership, not because passive investors privately learned news.

That said, index changes are not a perfect experiment:

- the committee does not choose additions randomly,
- firms added later in the sample differ from those added earlier,
- firms can migrate across indices,
- and some additions may have become partially predictable.

Interpretation. The paper's identification strategy is therefore layered rather than singular: event-study timing, composition controls, migration adjustment, and predictability checks all matter.

4.2 Why composition is unlikely to be the whole story

The paper directly controls for observable characteristics that should matter for price impact: turnover, size relative to the index, arbitrage risk, and analyst coverage. After these controls, the decade effects still show a large decline from the 1990s and 2000s to the 2010s.

Interpretation. This does not prove that all composition channels are ruled out, but it does show that the disappearing effect is not just a simple story about smaller, easier-to-arbitrage, or more widely followed firms entering the index.

4.3 Why migrations matter

The migration test is conceptually sharp. If a stock moves from the MidCap to the S&P 500, there are two offsetting passive trades. Once MidCap funds become large, the net demand shock should be much smaller than the raw S&P 500 demand shock.

The evidence lines up with this logic:

- the share of migration events rises over time,
- MidCap-linked assets rise sharply,
- migration returns become much smaller than direct-addition returns.

Interpretation. This is not merely a correlation. It is exactly the comparative static the demand-shock model predicts.

4.4 Why predictability is only a partial explanation

The paper has no position-level data on front runners, so this part is indirect. It looks at returns before the announcement and at how highly eventual additions rank by market capitalization.

The evidence is mixed:

- longer-run preannouncement returns are somewhat stronger in recent decades,
- the simple rule “pick the largest eligible firm” works somewhat better than before,
- but short-window preannouncement returns do not change much,
- and exact prediction of who gets added remains noisy.

Interpretation. The paper is careful here. It does not deny that predictability rose. It argues that predictability alone is too weak to explain the collapse in announcement-to-implementation returns.

4.5 Why the multiplier estimates are credible but not perfect

The multiplier calculations hinge on two assumptions:

- the measured net demand shock is a good proxy for true passive demand,
- and the relation between average return and average demand is approximately linear within decade.

These are strong simplifications, but the paper checks robustness by:

- separating migrations from direct events,
- adding stock controls,
- using both short and longer return windows,
- and showing complementary evidence from ownership and trading-cost data.

Interpretation. The multiplier should be read as a disciplined reduced-form summary of market elasticity, not as a structural deep parameter.

4.6 What the paper cannot fully identify

Two limitations are worth keeping in mind:

- There is no direct observation of arbitrageur inventories or front-running positions.

- The paper can document that other institutions absorb index demand, but it cannot fully pin down which institutions or contracts organize that liquidity in each episode.

Interpretation. The paper is strongest on the facts “the effect is gone” and “net shocks are increasingly absorbed.” It is somewhat less definitive on the exact microstructure channels doing the absorbing.

5 Tables and figures

5.1 Figure 1, Table I, and Figure 2: tracker demand rises while price impact vanishes

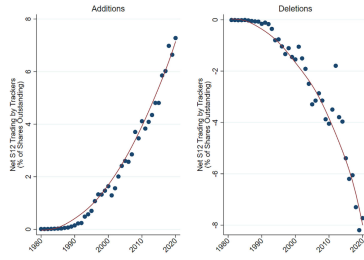


Figure 1. Net buying and selling by S&P 500 index trackers, by year. Net buying and selling are defined as the total change in split-adjusted shares held by index trackers between the quarter before and the quarter after the index change, divided by split-adjusted shares outstanding (multiplied by 100). Each point represents an equal-weighted average among events, by year. The red line represents a LOWESS filter with bandwidth equal to 0.8. (Color figure can be viewed at wileyonlinelibrary.com)

Table I
S&P 500 Addition and Deletion Cumulative Abnormal Returns by Year

Announcement returns (Ann.) are the cumulative abnormal returns from the day before to the day after the announcement. Effective returns (Eff.) are the cumulative abnormal returns from the day before to the day after the index change became effective. Total returns are the cumulative abnormal returns from the day before the announcement to the day after the index change became effective. The table reports means by year, as well as results pooled across each 10-year period (observations within each 10-year period are event-weighted). The second-to-last row of the table reports the difference in average returns between the 2000s and 2010s, while the last row of the table reports the associated standard errors (White (1980)). All returns are market-adjusted. Entries of “-” denote periods with no observations. In the last four rows, “*”, “**”, and “***” refer to significance at the 1%, 5%, and 10% level, respectively.

Additions Cumulative Abnormal Returns				Deletions Cumulative Abnormal Returns			
Year	N	Ann.	Eff.	Total	N	Ann.	Eff.
1980	5	5.90%	5.90%	5.90%	0		
1981	15	3.76%	3.76%	3.76%	2	-1.25%	-1.25%
1982	22	3.03%	3.03%	3.03%	4	-3.42%	-3.42%
1983	8	3.10%	3.10%	3.10%	6	-2.75%	-2.75%
1984	28	1.75%	1.75%	1.75%	3	-8.38%	-8.38%
1985	26	2.04%	2.04%	2.04%	2	-30.39%	-30.39%
1986	23	4.29%	4.29%	4.29%	3	8.73%	8.73%
1987	21	6.14%	6.14%	6.14%	2	-3.91%	-3.91%
1988	20	3.71%	3.71%	3.71%	5	-3.66%	-3.66%
1989	26	2.79%	3.33%	3.16%	12	-5.41%	-5.39%
1990	9	1.96%	2.01%	4.64%	4	-30.89%	-24.35%
1991	9	8.16%	5.12%	8.82%	4	-50.74%	-11.50%
1992	6	4.10%	3.39%	6.85%	0	-24.24%	-19.30%
1993	6	5.70%	5.28%	8.47%	3	-0.79%	-4.76%
1994	7	3.39%	1.02%	3.98%	6	-4.66%	0.27%
1995	14	2.97%	2.22%	5.67%	10	-5.91%	-7.51%
1996	16	4.67%	2.90%	7.88%	8	-6.32%	-2.65%
1997	14	6.41%	5.33%	10.07%	3	-8.32%	-1.25%
1998	23	5.19%	5.12%	9.21%	7	-9.92%	-6.68%
1999	30	5.43%	2.82%	6.48%	6	-16.14%	-6.71%
2000	35	7.21%	3.00%	9.51%	17	-13.83%	-5.12%
2001	22	3.78%	0.09%	5.49%	6	-8.81%	-11.27%
2002	15	3.72%	2.49%	6.35%	11	-9.08%	-9.99%
2003	7	2.54%	-0.16%	0.75%	3	-23.94%	-24.18%
2004	12	2.73%	2.01%	4.76%	6	-3.14%	-4.67%
2005	12	3.44%	0.93%	3.65%	2	-4.56%	-7.53%
2006	22	4.40%	1.77%	5.89%	6	-6.01%	-1.05%
2007	31	2.25%	1.47%	2.85%	4	2.41%	0.69%
2008	31	4.68%	1.41%	5.64%	14	-21.25%	-13.80%
2009	24	2.48%	-0.84%	1.50%	16	-3.52%	-2.68%
2010	13	2.31%	-1.49%	-0.59%	3	0.74%	4.34%
2011	8	0.17%	-1.00%	-1.87%	9	1.03%	0.25%
2012	5	4.61%	-1.33%	3.11%	5	-1.71%	-2.58%
2013	11	2.45%	1.54%	3.01%	10	-1.40%	2.95%
2014	8	1.79%	0.07%	0.68%	8	2.71%	-0.62%
2010s							
2000s							
vs.		-3.04%***	-1.09%***	-4.33%***		10.03%***	5.12%***
(SE)		(0.52)	(0.54)	(0.93)		(2.29)	(2.16)

(Continued)

Table I—Continued

Additions Cumulative Abnormal Returns				Deletions Cumulative Abnormal Returns			
Year	N	Ann.	Eff.	Total	N	Ann.	Eff.
2015	19	2.08%	0.46%	2.56%	6	-2.04%	-1.69%
2016	21	0.24%	-1.06%	-0.83%	7	4.18%	1.67%
2017	20	0.77%	0.18%	0.10%	12	1.46%	-0.33%
2018	21	0.36%	0.76%	-0.85%	7	-0.35%	-0.88%
2019	15	-0.70%	1.57%	0.47%	10	-6.87%	0.18%
2020	12	0.40%	2.31%	5.49%	10	1.11%	-0.77%
1990s	196	3.30%***	3.44%***	3.42%***	39	-4.71%***	-4.65%***
1990s	134	4.94%***	3.55%***	7.39%***	56	-13.91%***	-8.52%***
2000s	211	4.06%***	1.35%***	5.16%***	85	-10.21%***	-4.99%***
2010s	153	1.02%***	0.26%	0.83%	87	-0.18%	0.13%
2010s							
vs.		-3.04%***	-1.09%***	-4.33%***		10.03%***	5.12%***
(SE)		(0.52)	(0.54)	(0.93)		(2.29)	(2.16)

Read: Figure 1 shows that net buying and selling by S&P 500 trackers rises massively over time. Table I then shows that, despite this, the addition effect falls from 7.39% in the 1990s to 0.83% in the 2010s, while the deletion effect falls in magnitude from -16.11% to -0.62%. Figure 2 visualizes these decade-level declines using rolling windows.

Economic message. This is the central puzzle of the paper. Mechanical demand becomes larger, but price impact becomes smaller. The simple “more passive money means more price pressure” story is therefore incomplete.

5.2 Figure 3 and Table II: changing characteristics do not explain the decline

Read: Figure 3 shows there is no stable event-by-event relation between tracker purchases and returns. Table II then controls for turnover, firm size relative to index capitalization, arbitrage risk, and analyst coverage. Even after these controls, the decade coefficients still collapse: for additions, the fitted decade effect falls from about 7.4% in the 1990s to about 1.8% in 2010–2020 in the controlled specification; for deletions, the same decline remains very large.



Figure 2. Average index effect by year. For each event, we compute the cumulative abnormal return over the period of interest. Lines represent 10-year moving averages of the yearly averages, expressed in percentage points. (Color figure can be viewed at wileyonlinelibrary.com)

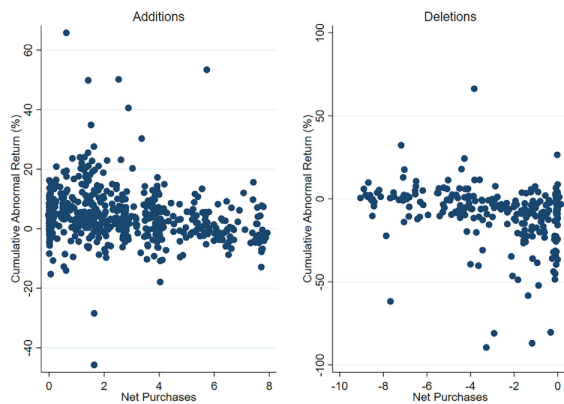


Figure 3. Net purchases and the index effect. The vertical axis represents the total inclusion effect return, that is, the abnormal return from the day before the announcement to the day after implementation, in percentage points. Each point represents an individual addition or deletion event. Net purchases are defined as the total change in split-adjusted shares held by index trackers between the quarter before and the quarter after the index change, divided by split-adjusted shares outstanding (multiplied by 100). (Color figure can be viewed at wileyonlinelibrary.com)

Table II
Cumulative Abnormal Returns, Controlling for Characteristics

We estimate the following multivariate regression separately for additions and deletions:

$$CAR_{it} = b_1 Turn_{i,t-1} + b_2 Size_{i,t-1} + b_3 WZ_{i,t-1} + b_4 Cover_{i,t-1} + \sum_{k=1}^4 \gamma_k 1_{era=k} + e_{it}$$

where CAR_{it} is the cumulative abnormal return from the day before the announcement date to the day after the implementation date, measured in percent. $Turn_{i,t-1}$ is average turnover (volume divided by shares outstanding) in stock i over the month before the index change, minus the value-weighted average turnover across all ordinary common shares traded on major exchanges in CRSP over the same period. $Size_{i,t-1}$ is the firm's market capitalization on the last day before the announcement of the index change relative to the total market capitalization of the S&P 500 on the same day. $WZ_{i,t-1}$ is Wurgler and Zhuravskaya's (2002) measure of arbitrage risk: the variance of CAPM regression residuals over the year before the announcement of index addition or deletion. $Cover_{i,t-1}$ is the number of analysts covering the stock during the last earnings announcement before the index change. All characteristics are demeaned. $1_{era=k}$ are dummy variables for 10-year periods. Robust standard errors are in parentheses (White (1980)). The rows labeled, for example, 2010s vs. 1980s, are differences in the coefficients between these two decades. p -Values from an F -test on equality of the γ_k coefficients are reported below the differences and are shown in square brackets. ***, **, and * refer to significance at the 1%, 5%, and 10% level, respectively.

	CAR Additions (%)		CAR Deletions (%)	
	(1)	(2)	(3)	(4)
$Turn_{i,t-1}$		-0.14 (0.49)		-0.637 (1.03)
$Size_{i,t-1}$		19.08*** (4.15)		28.45* (16.46)
$WZ_{i,t-1}$		3.868*** (1,173.00)		-2.183 (1,590.00)
$Cover_{i,t-1}$		-0.0289 (0.06)		0.0863 (0.17)
1980 to 1989	3.654*** (0.35)	4.594*** (0.58)	-2.184 (2.69)	-0.388 (3.20)
1990 to 1999	7.414*** (0.78)	7.388*** (0.86)	-15.20*** (2.40)	-13.09*** (2.34)
2000 to 2009	4.950*** (0.70)	4.592*** (0.61)	-12.71*** (2.37)	-9.517*** (1.80)
2010 to 2020	0.782 (0.55)	1.765*** (0.61)	-0.62 (1.12)	1.5 (1.96)
N	610	610	237	237
R^2	0.261	0.352	0.28	0.314
2010s vs. 1980s	-2.872 [0.000]	-2.829 [0.000]	1.564 [0.592]	1.889 [0.629]
2010s vs. 1990s	-6.632 [0.000]	-5.623 [0.000]	14.58 [0.000]	14.59 [0.000]
2010s vs. 2000s	-4.168 [0.000]	-2.827 [0.000]	12.09 [0.000]	11.02 [0.000]

Economic message. Bigger additions do generate bigger price effects, and composition matters somewhat, but the broad time-series collapse in the index effect survives the controls. So this is not mainly a composition story.

5.3 Figures 4–6 and Table III: migrations explain an important part of the decline

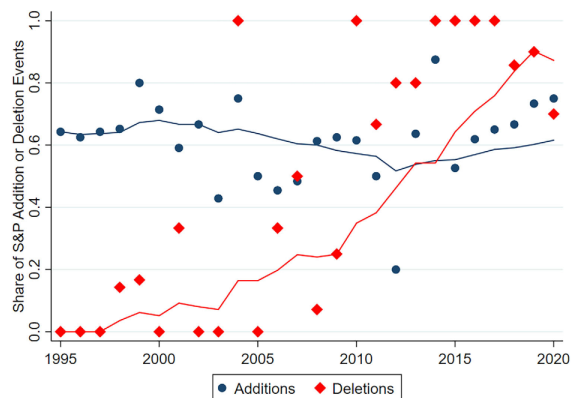


Figure 4. Migrations as a fraction of S&P 500 additions and deletions. Each year, we define migration additions as firms that are simultaneously added to the S&P 500 and dropped from the S&P MidCap, while migration deletions are firms that are simultaneously dropped from the S&P 500 and added to the S&P MidCap. Each data point represents the share of additions and deletions in our sample each year that are classified as migrations. Lines represent 10-year moving averages. (Color figure can be viewed at wileyonlinelibrary.com)

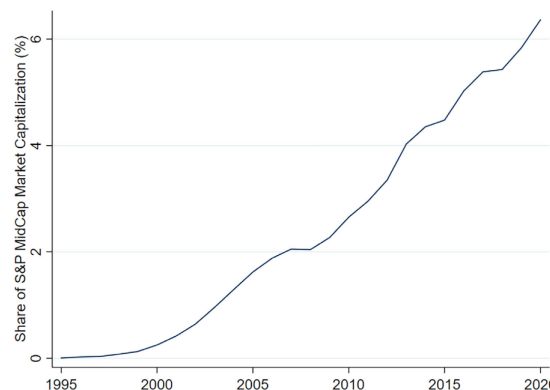


Figure 5. Percent of S&P MidCap 400 owned by index trackers. Each year, we identify S&P MidCap 400 index funds based on names and correlations. To identify funds based on names, we require that the fund name contain either variants of “S&P,” “SPDR,” or “S and P” as well as variants of “400” or “MidCap.” To identify funds based on correlations, we first restrict attention to the universe of midcap-focused equity funds (those with either CRSP objective code “EDCM” or a Lipper objective code that starts with MC) that do not include variants of “Vanguard” or “Russell” in the name. Among these funds, we classify them as S&P MidCap 400 trackers if their returns have a correlation of at least 99.5% with the index itself for three years in a row. Finally, we add up the total assets of these funds and divide them by the total market capitalization of the S&P MidCap 400 index. The line represents a 10-year moving average. (Color figure can be viewed at wileyonlinelibrary.com)

Read: Figure 4 shows that migration events become more common over time, especially for deletions. Figure 5 shows that S&P MidCap tracking also becomes large. Table III compares direct events and migrations. For additions, direct-addition CARs are 10.23% in 1995–1999, 8.80% in the 2000s, and 5.40% in the 2010s, while migration additions fall from 6.65% to 2.72% to –1.82%. Figure 6 plots the widening gap between the two.

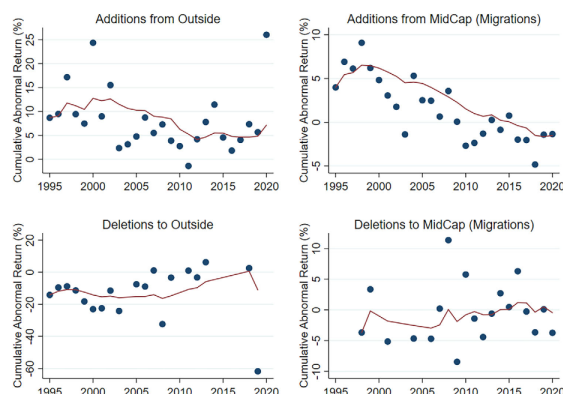


Figure 6. Comparing addition and deletion returns across index migrations and “direct additions” or “direct deletions.” For each event, we compute the abnormal return from the day before the announcement to the day after implementation. The dots represent the average abnormal return each year, expressed in percentage points. The lines represent 10-year moving averages. (Color figure can be viewed at wileyonlinelibrary.com)

Table III
Difference in Cumulative Abnormal Returns between Direct Additions and Migrations

This table presents mean cumulative abnormal returns from the day before the announcement to the day after implementation for migrations versus direct additions and deletions, as well as the average difference between each of these returns in each subperiod. Standard errors from two-sided *t*-tests are reported below the means.

		Additions CAR (%)			Deletions CAR (%)		
		Direct	Migrations	Diff.	Direct	Migrations	Diff.
1995 to 1999	Mean	10.23%	6.65%	–3.58%	–12.78%	–0.17%	12.61%
	(SE)	(7.69%)	(10.41%)	(2.12%)	(10.85%)	(4.98%)	(7.82%)
2000 to 2009	Mean	8.80%	2.72%	–6.07%	–15.71%	–4.10%	11.61%
	(SE)	(9.65%)	(9.29%)	(1.34%)	(25.61%)	(6.16%)	(6.29%)
2010 to 2020	Mean	5.40%	–1.82%	–7.22%	–6.86%	0.06%	6.92%
	(SE)	(8.38%)	(3.77%)	(0.99%)	(24.44%)	(8.45%)	(4.15%)

Economic message. Once MidCap funds become important, a stock moving from MidCap to S&P 500 no longer faces one-sided passive demand. This strongly reduces net buying pressure and

explains part of why average S&P 500 addition returns decline.

5.4 Figure 7, Table IV, and Figure 8: predictability rises a bit, but not enough

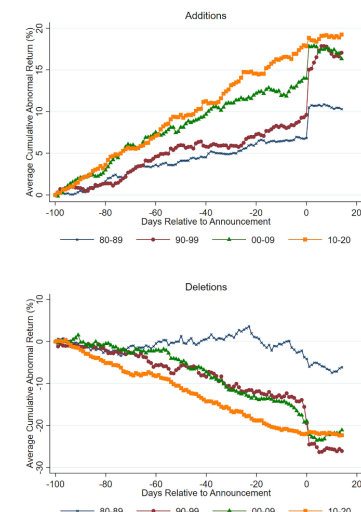


Figure 7. Cumulative abnormal preaddition and predeletion market-adjusted returns
The figure shows average cumulative abnormal returns in event time, pooled for 1980 to 1989, 1990 to 1999, 2000 to 2009, and 2010 to 2020. Event time of zero denotes the announcement date. Returns are normalized to start at zero 100-trading days before the announcement and are expressed in percentage points. (Color figure can be viewed at wileyonlinelibrary.com)

Table IV
Cumulative Abnormal Returns Including Longer Pre-Announcement
Windows

Announcement itself. Additions and deletions are shown separately						
		Mean CAR (%)				
		Time Relative to Announcement	1980 to 1989	1989 to 1990	1990 to 2000	2000 to 2010 to 2020
Cumulative abnormal returns: Additions	-100	-1	6.91%	9.60%	14.16%	18.61%
	-50	-1	2.71%	2.49%	5.66%	6.61%
	-20	-1	0.81%	1.56%	1.96%	2.56%
	-10	-1	0.39%	0.88%	1.31%	0.96%
	-5	-10	10.65%	17.21%	17.41%	19.79%
	-1	-10	6.37%	9.59%	9.57%	7.26%
Cumulative abnormal returns: Deletions	-100	-1	4.24%	8.19%	8.41%	3.01%
	-50	-1	3.89%	7.31%	4.80%	1.60%
	-20	-1	-2.81%	-11.68%	-20.10%	-22.33%
	-10	-1	-1.50%	-11.26%	-18.54%	-11.44%
	-5	-1	-5.55%	-12.51%	-13.15%	-10.95%
	-1	-10	-2.59%	-4.45%	-10.69%	-8.10%
Cumulative abnormal returns: Total	-100	-1	6.51%	9.257%	22.93%	22.16%
	-50	-1	6.60%	6.201%	22.32%	11.39%
	-20	-1	10.30%	15.65%	18.56%	10.66%
	-10	-1	-7.72%	-14.95%	-16.06%	-16.77%

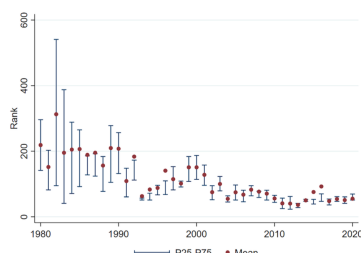


Figure 8. Rank of additions among stocks outside the S&P 500 index by year. Each month, we rank all ordinary common shares traded on major exchanges outside the S&P 500 by market capitalization. Then, each year, we plot the average rank as well as the 25th and 75th percentile ranks of stocks that ended up being added the month before their index addition. (Color figure can be viewed at wileyonlinelibrary.com)

Read: Figure 7 shows more of the total run-up for additions occurs before the announcement in recent decades. Table IV quantifies this: for additions, the $[-20, -1]$ abnormal return rises only from 1.54% in the 1990s to 2.35% in the 2010s; for deletions, the same short-window abnormal return becomes *less* negative, moving from -5.56% to -3.03% . Figure 8 shows eventual additions are more highly ranked by size than before, but the ranking range remains wide.

Economic message. The evidence is consistent with somewhat greater predictability, but not with a story in which front running fully shifts the index effect out of the announcement-to-implementation window. Exact inclusion choices are still hard to forecast.

5.5 Tables V and VI: the price-impact multiplier collapses

Table V
Index Effect Decomposition

This table represents estimates of multiplier M and elasticity $-1/M$ are based on

$$\hat{C}AR = M \times (\alpha \cdot \hat{\rho}_{\text{domestic}} + (1 - \alpha) \cdot \hat{\rho}_{\text{foreign}})$$

Return Measure	Sample	Period	N	CAR	$D_{\text{pre}}(p_{\text{CAR}})$	$D_{\text{post}}(p_{\text{CAR}})$	$\hat{\beta}_{\text{CAR}}$	$\hat{\beta}_{\text{CAR}}^{\text{boot}}$	$t = \pm 1$		
Return 1 day before announcement to 1 day after effective	Additive	0000-0000	211	0.75	0.71	0.20	0.20	1.61	3.38	-0.28	
		0000-0000	105	0.75	0.71	0.20	0.20	1.61	1.58	-0.28	
		0000-0000	106	0.75	0.71	0.20	0.20	1.61	1.58	-0.28	
	Differential	0000-0000	211	-11.6%	-0.44	-0.17	0.80	0.80	1.06	7.06	-0.11
		0000-0000	105	-12.4%	-0.52	-0.20	0.80	0.80	1.23	5.77	-0.17
		0000-0000	106	-12.4%	-0.52	-0.20	0.80	0.80	1.23	5.77	-0.17
Forecasted CAR 3 days before announcement to 1 day after effective	Additive	0000-0000	97	10.8%	1.13	1.30	0.90	0.90	1.33	0.86	-0.11
		0000-0000	49	10.8%	1.13	1.30	0.90	0.90	1.33	0.86	-0.11
		0000-0000	48	10.8%	1.13	1.30	0.90	0.90	1.33	0.86	-0.11
	Differential	0000-0000	97	-15.7%	-0.84	-1.07	0.85	0.85	1.06	14.74	-0.07
		0000-0000	49	-16.0%	-0.92	-1.20	0.85	0.85	1.23	12.05	-0.03
		0000-0000	48	-16.0%	-0.92	-1.20	0.85	0.85	1.23	12.05	-0.03

Table VI
Effect of Controls on Elasticity Estimates

This table presents estimates of the following multivariate regression separately for additions and deletions:

[illegible]

Table VI—Continued

	Asymmetric CAFE				Defective CAFE			
	Ann-1 to -EF-1		Ann-20 to -EF-1		Ann-1 to -EF-1		Ann-20 to -EF-1	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1995-1999 × Avg. demand shock	0.779***	0.664***	0.918***	0.715***	0.630***	0.793***	0.493***	0.105**
	(0.080)	(0.084)	(1.581)	(1.268)	(1.906)	(0.329)	(0.121)	(0.046)
2000-2009 × Avg. demand shock	0.841***	0.313***	0.869***	0.566***	0.566***	0.489***	0.813***	0.514**
	(0.097)	(0.071)	(0.671)	(0.658)	(1.043)	(0.085)	(3.286)	(1.997)
1990-2010 × Avg. demand shock	0.548	0.749***	1.167***	0.749***	0.606	(-0.183)	0.260*	0.040
	(0.245)	(0.275)	(0.574)	(1.125)	(0.235)	(0.315)	(0.235)	(0.235)
χ^2	451	451	451	451	266	266	266	266
df	0.031	0.04	0.202	0.503	0.207	0.277	0.344	0.625

Economic message. These are the paper’s most important quantitative results. Even after accounting for smaller net shocks due to migrations and after adding controls, the market absorbs a given demand shock far more easily than it used to.

5.6 Figure 9 and Table VII: who provides the liquidity?

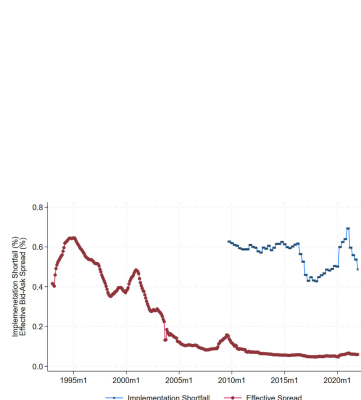


Figure 9. Value-weighted average effective bid-ask spreads and implementation shortfall. The percentage effective spread is computed using TAQ data and the method in Holden and Jacobson (2014). The red line plots the value-weighted percentage effective spread among all ordinary common shares traded on major exchanges, that is, the weights are proportional to each firm’s one-month lagged market capitalization. The effective spread is computed using second-based TAQ data from 1993 until millisecond TAQ data became available in 2003. Implementation shortfall—the blue line—is the difference, or slippage, between the arrival price and the execution price for a trade. The plot shows the average implementation shortfall from 2009 through 2021, using data from ITG and Virtu Financial, for midcap stocks. (Color figure can be viewed at wileyonlinelibrary.com)

Table VII
Net Buying and Selling by Trackers and Institutional Investors
This table presents net buying and selling by trackers and institutional investors defined as the total change in split-adjusted shares held by index trackers between the quarter before and the quarter after the index change, divided by split-adjusted shares outstanding (multiplied by 100). Net buying and selling by institutions (Insts.) is defined as the total change in split-adjusted shares held by 13F filing institutions between the quarter before and the quarter after the index change, divided by split-adjusted shares outstanding. Net buying by active and passive mutual funds is defined similarly. We compute the median of this quantity among additions and deletions each year. The final row represents an equal-weighted average across the yearly medians. Entries of “-” denote periods with no observations.

Year	Additions (Median by Year)			Deletions (Median by Year)		
	Trackers	Passive	Active	Trackers	Passive	Active
1980	0.01%	0.01%	0.41%	-0.18%		
1981	0.01%	0.01%	-0.03%	0.93%	-0.01%	0.54%
1982	0.01%	0.01%	0.35%	3.40%	-0.01%	0.18%
1983	0.02%	0.02%	0.11%	0.54%	-0.02%	-0.49%
1984	0.02%	0.02%	-0.14%	2.43%	-0.01%	-0.09%
1985	0.03%	0.03%	-0.25%	2.48%	-0.01%	-4.65%
1986	0.03%	0.03%	0.48%	3.31%	-0.02%	-2.52%
1987	0.05%	0.05%	0.14%	1.37%	-0.05%	3.75%
1988	0.06%	0.05%	0.42%	3.24%	-0.06%	-8.49%
1989	0.10%	0.07%	0.08%	2.22%	-0.07%	-18.13%
1990	0.15%	0.10%	1.14%	0.54%	-0.15%	-11.70%
1991	0.23%	0.19%	0.24%	1.55%	-0.12%	-1.11%
1992	0.28%	0.23%	0.33%	2.21%	-0.16%	-0.09%
1993	0.46%	0.34%	0.66%	1.38%	-0.41%	-1.18%
1994	0.57%	0.41%	-0.55%	0.77%	-0.66%	-0.40%
1995	0.68%	0.55%	0.60%	0.61%	-0.66%	-0.41%
1996	1.11%	0.95%	-0.66%	0.27%	-1.03%	-0.72%
1997	1.37%	1.12%	0.61%	0.27%	-1.29%	-0.70%
1998	1.47%	1.38%	0.14%	0.89%	-1.24%	-0.94%
1999	1.59%	1.49%	-1.01%	0.41%	-1.45%	-1.30%
2000	1.61%	1.58%	2.06%	1.36%	-1.56%	-0.80%
2001	1.08%	1.05%	-0.79%	0.42%	-0.91%	-0.93%
2002	1.36%	1.25%	0.01%	-0.06%	-1.19%	-1.23%
2003	2.13%	1.70%	-1.85%	-1.09%	-2.01%	-1.82%
2004	1.49%	1.40%	-0.79%	0.90%	-2.24%	-2.25%
2005	2.70%	1.43%	-0.74%	-0.20%	-3.29%	-3.17%
2006	2.63%	1.82%	-0.05%	0.36%	-3.07%	-2.08%
2007	2.80%	1.66%	0.46%	0.69%	-2.88%	-1.94%
2008	3.84%	2.99%	0.90%	1.55%	-3.45%	-2.46%
2009	2.76%	1.82%	0.25%	0.70%	-4.11%	-1.46%
2010	3.86%	1.57%	0.45%	1.01%	-4.28%	-1.09%
2011	4.01%	0.78%	1.11%	0.30%	-4.01%	-1.49%
2012	4.28%	2.65%	-4.49%	0.73%	-0.03%	0.33%
2013	4.69%	1.25%	0.09%	1.86%	-1.28%	0.71%
2014	4.87%	1.31%	0.07%	2.49%	-5.34%	-1.41%
2015	5.34%	2.12%	0.19%	1.11%	-6.32%	-0.90%
2016	5.99%	1.16%	0.41%	0.59%	-6.50%	-1.75%

(Continued)

Table VII—Continued

Year	Additions (Median by Year)			Deletions (Median by Year)		
	Trackers	Passive	Active	Trackers	Passive	Active
2017	6.35%	2.03%	2.46%	0.48%	-7.05%	-1.94%
2018	7.43%	0.88%	0.30%	0.45%	-7.68%	-0.31%
2019	7.57%	0.99%	-0.80%	1.16%	-8.46%	-2.35%
2020	7.53%	2.55%	2.40%	0.65%	-8.45%	-2.29%
Mean	3.04%	1.29%	0.13%	0.85%	-3.08%	-1.24%

Read: Figure 9 shows a sharp decline in trading costs: value-weighted effective spreads fall from roughly 60 bps to 6 bps, though implementation shortfall declines less. Table VII shows that in 2020, trackers buy 7.53% of shares outstanding for the average addition, but total institutional ownership rises only 0.65%. For deletions, trackers sell 8.45% while total institutional ownership falls only 1.42%. Active mutual funds are not the main counterparties.

Economic message. The market now has organized balance-sheet capacity to absorb index demand. The passive trade is no longer pushing against an empty book; other institutions, outside the tracker universe and often outside active mutual funds, are taking the other side.

5.7 Table VIII: similar pattern in other indices, but weaker

Read: Table VIII shows that addition and deletion effects in Russell, S&P MidCap, S&P SmallCap, and Nasdaq 100 often grow into the 2000s and then shrink in the 2010s. For example, Russell 2000

Table VIII
Addition and Deletion Returns by Year for Other Indices
This table presents mean cumulative abnormal returns (CAR) for additions to other indices. For direct additions to the Russell 2000 (R2 Addition) and Russell 1000 (R1 Addition), CAR_{it} is the cumulative market-adjusted return from the day before the ranking date in May to the day after implementation, typically in June. For the MidCap, SmallCap, and Nasdaq 100, only direct additions/deletions are included. In these cases, CAR_{it} is the cumulative market-adjusted return from 10 days before the effective date to one day after the effective date. Standard errors, clustered by year, are reported in parentheses. p -Values from an F -test of equality of the β coefficients is reported below the differences in the coefficients in square brackets. ***, **, and * refer to significance at the 1%, 5%, and 10% level, respectively.

	Russell		S&P MidCap		S&P SmallCap		Nasdaq 100		All	
	Addition	Addition	Addition	Deletion	Addition	Deletion	Addition	Deletion	Addition	Deletion
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1990s	1.011	2.215*	5.624***	-4.528**	6.342***	-6.694***	3.876	0.171	2.469*	-2.603***
	(1.317)	(1.238)	(1.591)	(1.777)	(1.032)	(0.640)	(3.276)	(1.537)	(1.275)	(0.203)
2000s	16.973*	8.284*	8.323***	9.352***	7.028***	-24.588***	5.262**	-2.822	8.226**	-16.711***
	(9.546)	(4.594)	(1.475)	(5.024)	(0.859)	(3.522)	(1.090)	(1.767)	(3.876)	(2.259)
2010s	8.532	3.148**	5.665***	-1.247	6.025***	-12.179***	1.976*	0.325	4.901***	-6.443***
	(5.114)	(1.421)	(1.125)	(2.478)	(0.735)	(1.606)	(1.140)	(0.676)	(1.286)	(0.502)
N	456	7,769	490	137	1,203	322	317	257	10,235	716
R ²	0.165	0.05	0.314	0.331	0.284	0.431	0.062	0.017	0.065	0.267
2000s vs. 1990s	15.962	0.069	2.709	-13.724**	0.684	-17.895***	-1.314	-2.793	5.767	-14.108***
[p-value]	[0.008]	[0.294]	[0.983]	[0.016]	[0.615]	[0.004]	[0.713]	[0.132]	[0.611]	[0.000]
2010s vs. 2000s	-8.441	-5.136	-2.668	17.005	-1.001	12.410***	-0.586	2.947	-4.225**	10.368***
[p-value]	[0.442]	[0.624]	[0.162]	[0.292]	[0.385]	[0.000]	[0.707]	[0.244]	[0.040]	[0.001]
2010s vs. 1990s	7.521	0.933	0.041	3.281***	-0.317	-5.485***	-1.900	0.154	1.542**	-3.840***
[p-value]	[0.165]	[0.212]	[0.223]	[0.006]	[0.805]	[0.004]	[0.589]	[0.928]	[0.023]	[0.000]

direct additions move from 2.2% in the 1990s to 8.3% in the 2000s and back to 3.1% in the 2010s. Pooled across other-index additions, the 2010s effect is 4.2 percentage points lower than the 2000s; pooled deletions are 10.3 percentage points less negative.

Economic message. The S&P 500 is the cleanest and strongest case, but the broad pattern is not unique. This supports the adaptive-markets interpretation rather than a one-off institutional quirk.

6 Short summary

- S&P 500 trackers' mechanical demand becomes much larger from 1980 to 2020, but addition and deletion abnormal returns nearly vanish by the 2010s.
- The paper frames the puzzle with $Price\ Impact = M \times D$: either the true shock D is smaller than it looks, or the multiplier M has fallen sharply.
- Changing composition of added and deleted firms explains only a limited portion of the decline.
- Migrations between the S&P MidCap and the S&P 500 are increasingly common and create offsetting passive flows, especially important for recent additions.
- Front running and predictability rise somewhat, but the evidence is too weak to explain the collapse in event-window returns on their own.
- After adjusting for migration-induced offsets, the implied price-impact multiplier falls by roughly a factor of 20 for additions and even more for deletions.
- The market's greater elasticity seems to reflect deeper liquidity, more specialized index-event trading, and institutional investors willing to absorb passive order flow.
- Other index families show a similar broad decline, though with weaker statistical precision than the S&P 500.
- The paper's bottom line is that a famous demand-based anomaly has been arbitrated away to a large extent.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the classic S&P 500 index effect has largely disappeared in the modern era, despite much larger passive demand shocks. This is not mainly because index changes no longer matter mechanically, nor because the firms entering the index are completely different. Instead, it is because the market has adapted.

7.2 Economic intuition

In the early decades of indexing, passive demand shocks were unusual and the market had limited specialized capital ready to absorb them. As indexing grew, the shocks became both larger and

more predictable. That created an incentive for liquidity providers, trading desks, and institutional counterparties to organize around rebalancing events. The result is a market that is much more elastic to repeated, well-understood demand shocks than it once was.

7.3 Takeaway

This paper is an important reminder that demand-based anomalies are not fixed objects. Once they become salient and tradable, markets can redesign themselves around them. The disappearance of the index effect therefore supports an adaptive view of market efficiency: repeated nonfundamental demand shocks need not generate persistent price pressure forever.